Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Shreya Anthey	anthey@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Goal: Teach civil engineering through a game-like experience

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

ASCE Bridge Designer, Bridge Builder and Truss Simulator, Bridge Designer - all websites found on Google

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

The functionality of my website would be to simulate how it is to create an actual bridge drivers can cross. It’d be used to predict whether or not the bridge “passes” or “fails”

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal my program will accomplish for the user is to get a point-of-view of how it is for a civil engineer designing a bridge that'd support transportation (cars, trucks, etc.)

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Some specific info needed to be provided in the program would be the location of a busy city; a budget of $600,000; the number of construction workers

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It'd calculate the amount of materials based on the budget, number of construction workers, and potentially even the time duration of when it is completed

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user would get back the total budget after calculations and what day/month the bridge would officially be completed

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Website for bridge building simulation: “Construct”, “Brainstorm”, “Coordinate”, “Design”, “Site plan”, “Location”, “Price”

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Most important “thing”: “Construct” - details I’ll need: location, cost, price, length ; details I’ll ignore: color, type

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

It’s most likely going to be calculating the budget and what’s leftover as well as the length of the bridge itself

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will understand the generated code with an LLM by writing it down through handwritten notes to go back and figure out how to go about with the codes I need to do so that way I understand it easier

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

The ultimate goal for this project is to express my creativity through a civil engineer's POV and get a full grasp as to how it'd be in the actual profession

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

So I'm doing a project right now regarding making a simple website to have people play an interactive game related to what it's like building a bridge like how a civil engineer would. What're 3 problems civil engineers have faced when planning out establishing bridges, and what're 3 simple features a stable bridge should have to support transportation on a daily basis?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

Regarding real-world problems civil engineers face with making bridges: difficult terrain & environmental conditions, supporting weight & distributing loads, as well as budget & material limitations; as for 3 importation features: strong foundations & supports, strong structural framework; and a safe yet durable roadway

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Ok, I notice that for the 1st problem you mentioned regarding what civil engineers faced, it's a bit too broad, I'm talking about simulating making a bridge in the location of a bustling city,

so can you focus in on that a bit more rather than being broad?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

It definitely gave me a lot more insight regarding how it'd be constructing a bridge in an already built urban setting

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I'd continue to use an LLM or visit office hours because I would like support in case I get stuck anywhere in creating my website

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Trace back to see what I did wrong and hopefully debug it on my own or get help from someone who has better knowledge than me

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

So I'm doing a project right now regarding making a simple website to have people play an interactive game related to what it's like building a bridge like how a civil engineer would. What're 3 problems civil engineers have faced when planning out establishing bridges, and what're 3 simple features a stable bridge should have to support transportation on a daily basis? That's a really interesting project idea! You could make your website feel like a mini civil engineering simulator where players have to make decisions about bridge construction

and see whether their designs can withstand real-world challenges.
Let's break it into two categories.
1. Three problems civil engineers face when planning bridges

1. Difficult terrain and environmental conditions

Bridges sometimes need to cross deep rivers, valleys, mountains, or unstable ground. Engineers must figure out where and how to build without the environment damaging the structure.

Game idea: Give players different locations and ask them to choose the best building site.

2. Supporting weight and distributing loads

Bridges must support cars, trucks, pedestrians, and sometimes much heavier vehicles. Engineers must calculate how much weight the bridge can handle and distribute that weight safely.

Game idea: Have players add vehicles to their bridge and see if it can handle the total weight.

3. Budget and material limitations

Engineers must work within a construction budget while choosing materials such as steel, concrete, and timber. Some materials are stronger or more durable but cost more.

Game idea: Give players a limited budget and let them decide which materials to purchase.

2. Three important features of a stable bridge

1. Strong foundations and supports

The foundation transfers the bridge's weight into the ground. Strong supports help prevent the bridge from sinking, shifting, or collapsing.

2. A strong structural framework

Bridges need a framework that distributes weight throughout the structure. Triangular trusses, beams, and cables can help distribute forces and prevent excessive bending.

3. A safe and durable roadway

A bridge needs a stable driving surface, protective guardrails, and materials that can withstand weather, repeated traffic, and everyday wear.

3. How you could turn these into an interactive game

Here is one possible game structure:

Game stage	Player's challenge
Stage 1: Location	Choose where to build the bridge.
Stage 2: Materials	Select steel, concrete, or wood.
Stage 3: Design	Choose supports and structural features.
Stage 4: Testing	Send vehicles across the bridge.
Stage 5: Results	See whether the bridge survives the test and stays within budget.

4. A few questions to think about for your website

Since the goal is to make an interactive game, think about what information your program needs to represent.

For example, your textbook's Values reading explains that abstraction means representing the real world as data by ignoring details. You don't need to simulate every detail of real bridge construction. You can simplify it into things like bridge strength, material cost, and vehicle weight.

Think about these questions:

1. Should players receive a limited amount of money to build their bridge?
2. Should different vehicles have different weights?
3. Should the bridge receive a score based on its strength, safety, and cost?

My question for you: Are you planning to make the game more like a building simulator (where players physically place bridge pieces), or more like a decision-making game (where players choose materials and designs from options)?

That distinction will help determine how to organize your game's features!

Ok, I notice that for the 1st problem you mentioned regarding what civil engineers faced, it's a bit too broad, I'm talking about simulating making a bridge in the location of a bustling city, so can you focus in on that a bit more rather than being broad?

Absolutely! If your game takes place in a bustling city, the first problem should focus on the challenges of building a bridge in an already-developed urban area.

Problem 1: Limited space and existing city infrastructure

Civil engineers often have to plan bridge construction around existing buildings, roads, sidewalks, and underground utilities (such as water pipes, gas lines, and electrical cables). They also need to avoid disrupting traffic and the daily routines of people living and working nearby.

How this could work in your game:

- The player chooses where to build the bridge, but nearby buildings and roads limit the available space.
- The player must avoid underground utility lines when selecting foundation locations.
- The player must keep traffic moving while construction takes place.

A game challenge: The player has to find a suitable bridge location without blocking too many roads or interfering with nearby buildings.

This makes the problem much more specific to a city setting than simply dealing with difficult terrain.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.